TIMOLOL

Systemic β-blocker effects

Punctal occlusion

CHEMICAL NAME (S)-1-(tert-butylamino)-3-[(4-morpholin-4-yl-1,2,5-thiadiazol-3-yl)oxy]propan-2-ol (maleate)

CLASS Non-selective β-blocker (ophthalmic antiglaucoma)

TRADE NAMES Timoptol · Timoptic · Cosopt (with dorzolamide) · Xalacom (with latanoprost)

THERAPEUTIC USES / INDICATIONS

- Open-angle glaucoma and ocular hypertension.
- Infantile haemangioma (topical gel, off-label).

PHARMACODYNAMICS

- Blocks **β-receptors on ciliary body** → ↓ aqueous humour production → ↓ intraocular pressure.

ADMINISTRATION

- **Eye drops:** 1 drop into affected eye(s); then close eyes and press inner corner (punctal occlusion) for 1–2 min to reduce systemic absorption.
- Gel-forming solution once daily.
- Wait 5–10 min between different eye drops; remove soft contact lenses (benzalkonium).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Stinging, burning, dry eye.
- Blurred vision.
- Fatigue.

ADVERSE EFFECTS (SERIOUS)

- Bronchospasm (asthma/COPD); fatal cases reported.
- Bradycardia, heart block, hypotension, heart failure.
- Masked hypoglycaemia; depression; reduced exercise tolerance.

DOSAGES

- **0.25–0.5%:** 1 drop twice daily.
- **Gel-forming (0.1–0.5%):** 1 drop once daily (morning).

CONTRAINDICATIONS & CAUTIONS

- Asthma or history of bronchospasm; severe COPD.
- Sinus bradycardia, 2nd/3rd-degree AV block, cardiogenic shock, uncontrolled heart failure.

KEY INTERACTIONS

- **Oral β-blockers:** additive effects.
- **Verapamil, diltiazem, digoxin:** bradycardia.
- **Quinidine, SSRIs (CYP2D6 inhibitors):** ↑ systemic levels.

PHARMACOKINETICS

- **Onset:** 20–30 min; max effect 1–2 h. **Duration:** 12–24 h.
- Systemically absorbed via nasolacrimal duct (no first-pass) → systemic β-blockade.

NURSING MANAGEMENT

- Ask about asthma, COPD, heart disease before starting.
- Teach drop instillation and punctal occlusion.
- Monitor pulse; IOP follow-up.

PATIENT EDUCATION

- After each drop, close your eye and press the inner corner for 1–2 minutes.
- Report wheeze, breathlessness, slow pulse, dizziness.
- Use regularly; glaucoma has no symptoms until vision is lost.

KEY POINTS

- Ophthalmic β-blocker — ↓ aqueous production.
- Systemic effects: bronchospasm, bradycardia.
- 1 drop BD (0.25–0.5%).
- Contraindicated in asthma.